

K Shiv Kumar (Shiv)

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Software/Systems Engineer with a strong foundation in networking and low-level systems, combining **1.8 years** of experience in industry with **more than 4 years** of research in programmable data planes (P4 and eBPF) and compiler framework (LLVM). Experienced in C/C++, Python, Linux Networking Stack, Host Networking, Linux Operating System, and Systems programming, with a proven ability to build solutions for real-world systems challenges.

EDUCATION

IIT Hyderabad - (NETX Lab) Kandi, Telangana
MS (Computer Science and Engineering) - (GPA: 7.60) 2021–2026
Research: eBPF, P4 and Compilers

IIIT Naya Raipur Naya Raipur, Chhattisgarh
MTech (Computer Science and Engineering) - (GPA: 8.78) 2019–2021

Shri Shankaracharya Engineering College Bhilai, Chhattisgarh
B.E. (Information Technology) - (74.48%) 2011–2015

SKILLS

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|----------------|--------------------|---------------------|-----------------------|
| • C/C++/Python | • Computer Network | • Shell Scripting | • HTML/CSS/SQL |
| • eBPF(XDP/TC) | • TCP/IP Stack | • Virtualization | • GitHub |
| • P4 Language | • SDN Concepts | • Docker/Kubernetes | • Systems Programming |
| • LLVM/Clang | • Linux Networking | • AWS (Beginner) | • Tech. Documentation |

PROJECTS

- “Yaksha-Prashna: Understanding eBPF Bytecode Network Function Behaviour” | **ACM CoNEXT '2026:** (🔗)
- Developed a system to validate the correctness and interactions of third-party eBPF network function bytecode.
 - Applied program analysis to extract functional properties of eBPF network functions from the bytecode.
 - Tools & Frameworks:** C++, STL library, LLVM, eBPF/XDP, Libbpf, Linux kernel, bpftool, bpfman.
- “In-Network Probabilistic Monitoring Primitives under the Influence of Adversarial Network Inputs” | **APNET '23:** (🔗)
- Analyzed vulnerabilities of monitoring primitives in programmable data planes against adversarial inputs.
 - Demonstrated FlowRadar pollution attacks where a few malicious flows corrupted telemetry data.
 - Quantified attack impact, revealing a 99% drop in FlowRadar’s accuracy under targeted adversarial conditions.
 - Tools & Frameworks:** Python, Wireshark, MS Excel.
- “DBVal: Validating P4 Data Plane Runtime Behaviour” | **ACM SOSR '21:** (🔗)
- Developed DBVal, a runtime validation system to detect packet-processing errors in P4 data planes.
 - Implemented assertion-based tracking of tables and actions, enabling line-rate validation of runtime Behaviour.
 - Tools & Frameworks:** Python, P4 language, Lex & Yacc, BMv2, Mininet, Wireshark, VirtualBox.

PROFESSIONAL EXPERIENCE

- Application Development Analyst (Accenture, Pune, India)** Mar 2016–Dec 2017
- Integrated a Learning Management System and developed an off-boarding feature for the client’s Salesforce environment, enhancing user onboarding and lifecycle workflows.
 - Award:** Apex Award - Extra Miller, in Accenture India.

CERTIFICATIONS

- Intel Connectivity Academy Level 1** | Issued by: Intel (🔗)

AWARDS AND HONORS

- ACM SIGMETRICS Student grant: For presenting our ongoing research on “Detecting Adversarial Attacks on Bloom Filters in P4 Data Plane Systems” (2022). (🔗)
- COMSNETS Travel Grant: For attending the Conference (2022). (🔗)